Infinitely Many Integer Solutions of $x^3 + x^2y^2 + z^2 + 1 = 0$

Abstract

We prove, unconditionally and by an entirely elementary argument, that

$$x^3 + x^2y^2 + z^2 + 1 = 0$$

has infinitely many integer solutions. The construction combines a norm identity in the Gaussian integers with an explicitly verified polynomial identity. A self-contained, congruence-controlled Pell argument then shows that an associated quadratic polynomial assumes square values at infinitely many integer arguments. Every numerical identity used in the construction is checked coefficient by coefficient.

1 Statement and preliminary reduction

We establish the following result.

Theorem 1.1 (Main theorem)**.** *The Diophantine equation*

$$x^3 + x^2y^2 + z^2 + 1 = 0 \tag{1}$$

has infinitely many pairwise distinct solutions $(x, y, z) \in \mathbb{Z}^3$.

A solution of (1) necessarily has $x < 0$, since all four terms on the left are nonnegative when $x \geq 0$. Writing

$$a = -x > 0,$$

we obtain the equivalent equation

$$z^2 + 1 = a^2(a - y^2). \tag{2}$$

In particular, every solution satisfies $a > y^2$. Our task is therefore to construct infinitely many triples (a, y, z) satisfying (2).

The signs of y and z are immaterial in (1). Once one solution is found, the four sign choices $(x, \pm y, \pm z)$ are again solutions. We shall nevertheless prove infinitude already at the level of the x-coordinates.

2 A Gaussian norm construction

The basic algebraic mechanism is the following lemma.

Lemma 2.1 (Gaussian norm mechanism)**.** *Let* $P, Q, R, S, Y \in \mathbb{Z}$ *satisfy*

$$2PQR + (P^2 - Q^2)S = 1, \tag{3}$$

$$P^2 + Q^2 - R^2 - S^2 = Y^2. \tag{4}$$

Define

$$X = -(P^2 + Q^2), \tag{5}$$

$$Z = (P^2 - Q^2)R - 2PQS. \tag{6}$$

Then

$$X^3 + X^2Y^2 + Z^2 + 1 = 0.$$

Proof. Put

$$A_0 = P^2 - Q^2, \qquad B_0 = 2PQ.$$

Condition (3) says that

$$B_0R + A_0S = 1,$$

whereas (6) says that

$$Z = A_0R - B_0S.$$

The elementary two-square identity gives

$$\begin{aligned} Z^2 + 1 &= (A_0R - B_0S)^2 + (B_0R + A_0S)^2 \\ &= (A_0^2 + B_0^2)(R^2 + S^2). \end{aligned}$$

Now

$$A_0^2 + B_0^2 = (P^2 - Q^2)^2 + 4P^2Q^2 = (P^2 + Q^2)^2.$$

Writing $a = P^2 + Q^2$, condition (4) becomes

$$R^2 + S^2 = a - Y^2.$$

Consequently,

$$Z^2 + 1 = a^2(a - Y^2). \tag{7}$$

Condition (3) also shows that P and Q cannot both vanish, so $a > 0$. Since $X = -a$, equation (7) yields

$$\begin{aligned} X^3 + X^2Y^2 + Z^2 + 1 &= -a^3 + a^2Y^2 + a^2(a - Y^2) \\ &= 0. \end{aligned}$$

This proves the lemma. $\square$

Equivalently, lemma 2.1 is the norm identity obtained from

$$(P + iQ)^2(R + iS) = Z + i.$$

We shall not need any theory of Gaussian integers beyond the explicitly expanded identity above.

3 The polynomial identities

For $t \in \mathbb{Z}$, define four integral linear polynomials

$$P(t) = 31487t + 15586, \tag{8}$$
$$Q(t) = -36963t - 18295, \tag{9}$$
$$R(t) = -3400t - 1682, \tag{10}$$
$$S(t) = 21114t + 10451. \tag{11}$$

The following proposition records the exact identities that make the construction work.

Proposition 3.1 (Explicit polynomial identities)**.** *For every integer t,*

$$2P(t)Q(t)R(t) + (P(t)^2 - Q(t)^2)S(t) = 1, \tag{12}$$

and

$$P(t)^2 + Q(t)^2 - R(t)^2 - S(t)^2 = F(t), \tag{13}$$

where

$$F(t) = 1900333542t^2 + 1881226506t + 465577896. \tag{14}$$

Proof. The verification is a direct calculation. To make every cancellation auditable, we list the coefficient of each power of t.

For (12), the two cubic polynomials on the left have the following coefficients:

power	$[t^j]\,2PQR$	$[t^j]\,(P^2 - Q^2)S$	sum
$j = 3$	7914207070800	-7914207070800	0
$j = 2$	11749892676484	-11749892676484	0
$j = 1$	5814858098812	-5814858098812	0
$j = 0$	959230706680	-959230706679	1

Thus every nonconstant coefficient cancels and the constant coefficient is 1, proving (12).

For (13), the corresponding quadratic coefficients are

power	$[t^j]\,(P^2 + Q^2)$	$[t^j]\,(R^2 + S^2)$	difference
$j = 2$	2357694538	457360996	1900333542
$j = 1$	2333988934	452762428	1881226506
$j = 0$	577630421	112052525	465577896

This proves (13) and (14) coefficient by coefficient. □

Indeed, if $F(t) = y^2$ for some integers t and y, then (12) and (13) are exactly the two hypotheses of lemma 2.1. The problem is therefore reduced to proving that $F(t)$ is a square for infinitely many integers t. We give a fully self-contained Pell-type argument.

4 A self-contained Pell propagation theorem

For a fixed positive nonsquare integer D and an element $\alpha = u + v\sqrt{D}$, define

$$\overline{\alpha} = u - v\sqrt{D}, \qquad \mathrm{N}(\alpha) = \alpha\overline{\alpha} = u^2 - Dv^2.$$

Direct multiplication gives $\overline{\alpha\beta} = \overline{\alpha}\,\overline{\beta}$ and hence

$$\mathrm{N}(\alpha\beta) = \mathrm{N}(\alpha)\,\mathrm{N}(\beta). \tag{15}$$

We first prove the only existence fact about Pell equations that will be used.

Lemma 4.1 (Existence of a nontrivial Pell unit)**.** *Let $D > 0$ be a nonsquare integer. Then there exist integers $r > 1$ and $s > 0$ such that*

$$r^2 - Ds^2 = 1. \tag{16}$$

Proof. Since D is a positive nonsquare integer, $\sqrt{D}$ is irrational. For each integer $N \geq 2$, consider the $N + 1$ fractional parts

$$\{0\sqrt{D}\}, \{1\sqrt{D}\}, \ldots, \{N\sqrt{D}\}$$

in the interval $[0, 1)$. Partition $[0, 1)$ into N half-open intervals of length $1/N$. By the pigeonhole principle, two of the fractional parts lie in the same interval. Subtracting them, we obtain integers p and q such that

$$1 \leq q \leq N, \qquad |q\sqrt{D} - p| < \frac{1}{N}. \tag{17}$$

For all sufficiently large N, the integer p is positive.

Infinitely many distinct pairs (p, q) occur in this way. Indeed, if only finitely many occurred, the finitely many positive numbers $|q\sqrt{D} - p|$ would have a positive minimum; this would contradict (17) for sufficiently large N.

For every such pair, put

$$k = p^2 - Dq^2.$$

The integer k is nonzero, because D is not a square and $q \neq 0$. Moreover, using (17), $q \leq N$, and $p < q\sqrt{D} + 1/N$, we obtain

$$\begin{aligned}
|k| &= |p - q\sqrt{D}|\,(p + q\sqrt{D}) \\
&< \frac{1}{N}\left(2N\sqrt{D} + 1\right) \\
&< 2\sqrt{D} + 1.
\end{aligned}$$

Thus k belongs to a fixed finite set of nonzero integers. One value of k must therefore occur for infinitely many distinct positive pairs (p, q).

Among those infinitely many pairs, choose two distinct pairs (p_1, q_1) and (p_2, q_2) that are componentwise congruent modulo $|k|$. This is possible because there are only $|k|^2$ residue classes. Define

$$r_0 = \frac{p_1p_2 - Dq_1q_2}{k}, \qquad s_0 = \frac{q_1p_2 - p_1q_2}{k}. \tag{18}$$

The congruences imply that both numerators in (18) are divisible by k, so $r_0, s_0 \in \mathbb{Z}$. Multiplying in $\mathbb{Z}[\sqrt{D}]$ gives

$$r_0 + s_0\sqrt{D} = \frac{(p_1 + q_1\sqrt{D})(p_2 - q_2\sqrt{D})}{k}.$$

Taking norms therefore yields

$$r_0^2 - Ds_0^2 = \frac{(p_1^2 - Dq_1^2)(p_2^2 - Dq_2^2)}{k^2} = 1.$$

We also have $s_0 \neq 0$. Otherwise $q_1 p_2 = p_1 q_2$, so the two positive pairs are positive rational multiples of one another. Since both have the same nonzero norm k, the square of that multiplier is 1, forcing the multiplier to be 1 and the two pairs to be equal, a contradiction.

Finally, set $r = |r_0|$ and $s = |s_0|$. Then $s > 0$ and $r^2 = 1 + Ds^2 > 1$, so $r > 1$. Equation (16) follows. $\square$

We next arrange the congruences needed to return from a generalized Pell equation to an integral value of the original variable.

Lemma 4.2 (Congruence-controlled Pell unit). *Let $D > 0$ be a nonsquare integer and let $M \geq 1$. There exist integers $\rho > 1$ and $\sigma > 0$ such that*

$$\rho^2 - D\sigma^2 = 1, \tag{19}$$

$$\rho \equiv 1 \pmod{M}, \tag{20}$$

$$\sigma \equiv 0 \pmod{M}. \tag{21}$$

Proof. Choose $r > 1$ and $s > 0$ as in lemma 4.1. If $M = 1$, the congruence conditions are vacuous, and $(\rho, \sigma) = (r, s)$ is sufficient. Hence assume $M \geq 2$ and put

$$\varepsilon = r + s\sqrt{D}.$$

Consider the finite ring

$$\mathcal{R}_M = (\mathbb{Z}/M\mathbb{Z})[w]/(w^2 - D).$$

Because the defining polynomial is monic of degree 2, every class has a unique representative $a + bw$ with $a, b \in \mathbb{Z}/M\mathbb{Z}$; in particular, $\mathcal{R}_M$ has exactly M^2 elements. The residue class of ε is a unit in $\mathcal{R}_M$, because its inverse is the residue class of $r - s\sqrt{D}$ and

$$(r + s\sqrt{D})(r - s\sqrt{D}) = 1.$$

The unit group of the finite ring $\mathcal{R}_M$ is finite. Hence the class of ε has finite multiplicative order: for some $h \geq 1$,

$$\varepsilon^h \equiv 1 \pmod{M}$$

coefficientwise in the basis $1, \sqrt{D}$. Write

$$\varepsilon^h = \rho + \sigma\sqrt{D}.$$

Then (20) and (21) hold. Norms are multiplicative, so (19) holds as well. Finally, because $r > 1$ and $s > 0$, every positive power of $r + s\sqrt{D}$ has positive integral coefficients; hence $\rho > 1$ and $\sigma > 0$. $\square$

We can now prove the square-value theorem needed later.

Proposition 4.3 (Pell propagation for quadratic square values)**.** *Let*

$$f(T) = AT^2 + BT + C \in \mathbb{Z}[T],$$

where $A > 0$ is not a square. Assume that

$$\Delta = B^2 - 4AC \neq 0 \tag{22}$$

and that there is at least one pair $(T_0, Y_0) \in \mathbb{Z}^2$ satisfying

$$Y_0^2 = f(T_0). \tag{23}$$

Then $Y^2 = f(T)$ has infinitely many integer solutions (T, Y). More precisely, the resulting integer values of T are unbounded in absolute value.

Proof. Completing the square shows that (23) is equivalent to

$$U_0^2 - AV_0^2 = \Delta, \tag{24}$$

where

$$U_0 = 2AT_0 + B, \qquad V_0 = 2Y_0. \tag{25}$$

Indeed,

$$(2AT + B)^2 - A(2Y)^2 = B^2 + 4A(AT^2 + BT - Y^2),$$

and this equals $B^2 - 4AC$ precisely when $Y^2 = AT^2 + BT + C$.

Apply lemma 4.2 with $D = A$ and $M = 2A$. We obtain $\rho > 1$ and $\sigma > 0$ such that

$$\rho^2 - A\sigma^2 = 1, \qquad \rho \equiv 1 \pmod{2A}, \qquad \sigma \equiv 0 \pmod{2A}. \tag{26}$$

For $n \geq 0$, define $U_n, V_n \in \mathbb{Z}$ by

$$U_n + V_n\sqrt{A} = (U_0 + V_0\sqrt{A})(\rho + \sigma\sqrt{A})^n. \tag{27}$$

Equivalently, they satisfy the integral recurrence

$$U_{n+1} = \rho U_n + A\sigma V_n, \tag{28}$$

$$V_{n+1} = \sigma U_n + \rho V_n. \tag{29}$$

Taking norms in (27) and using (26) gives

$$U_n^2 - AV_n^2 = \Delta \qquad (n \geq 0). \tag{30}$$

The congruences in (26), together with (28) and (29), imply inductively that

$$U_n \equiv U_0 \pmod{2A}, \qquad V_n \equiv V_0 \pmod{2A}. \tag{31}$$

Since $U_0 \equiv B \pmod{2A}$ and V_0 is even, the quantities

$$T_n = \frac{U_n - B}{2A}, \qquad Y_n = \frac{V_n}{2} \tag{32}$$

are integers. Substituting (32) into (30), and reversing the completed-square calculation, yields

$$Y_n^2 = AT_n^2 + BT_n + C = f(T_n).$$

It remains to prove that infinitely many of these solutions are distinct.

Set

$$\alpha = U_0 + V_0\sqrt{A}, \qquad \varepsilon = \rho + \sigma\sqrt{A} > 1.$$

Because $\mathrm{N}(\alpha) = \Delta \neq 0$, we have $\alpha \neq 0$. The conjugate of ε is ε^{-1}, since $\mathrm{N}(\varepsilon) = 1$. Hence (27) and its conjugate give

$$U_n = \frac{\alpha\varepsilon^n + \overline{\alpha}\,\varepsilon^{-n}}{2}. \tag{33}$$

The first term on the right of (33) tends to infinity in absolute value, while the second tends to 0. Thus $|U_n| \to \infty$, and therefore $|T_n| \to \infty$ by (32). In particular, the pairs (T_n, Y_n) include infinitely many distinct integer solutions. □

Remark 4.4. The hypothesis $\Delta \neq 0$ in proposition 4.3 is used only to ensure that the initial generalized-Pell element $U_0 + V_0\sqrt{A}$ is nonzero, which makes the orbit unbounded. In the application below, the discriminant is explicitly negative and nonzero.

5 Infinitely many square values of the auxiliary quadratic

We apply proposition 4.3 to the polynomial F in (14). Set

$$A = 1900333542, \qquad B = 1881226506, \qquad C = 465577896. \tag{34}$$

First, A is not a square, because

$$43592^2 = 1900262464 < 1900333542 < 1900349649 = 43593^2. \tag{35}$$

Second, the discriminant is

$$B^2 - 4AC = -1853382492 \neq 0. \tag{36}$$

Third, F has the following integral square value:

$$\begin{aligned} F(-659) &= 1900333542 \cdot 659^2 - 1881226506 \cdot 659 + 465577896 \\ &= 825278750953302 - 1239728267454 + 465577896 \\ &= 824039488263744 \\ &= 28706088^2. \end{aligned} \tag{37}$$

All hypotheses of proposition 4.3 are therefore satisfied.

Corollary 5.1. *There exist infinitely many pairs $(t, y) \in \mathbb{Z}^2$ satisfying*

$$y^2 = F(t). \tag{38}$$

Moreover, the corresponding integers t are unbounded in absolute value.

Proof. Apply proposition 4.3 using (34), (35), (36), and (37). □

For completeness, the Pell orbit in this application begins with

$$\begin{aligned} U_0 &= 2A(-659) + B = -2502758381850, \\ V_0 &= 2 \cdot 28706088 = 57412176, \end{aligned}$$

and it satisfies

$$U_0^2 - AV_0^2 = -1853382492.$$

Any norm-one Pell unit $\rho + \sigma\sqrt{A}$ satisfying

$$\rho \equiv 1 \pmod{2A}, \qquad \sigma \equiv 0 \pmod{2A}$$

then gives the explicit recurrence (28), (29), and (32). The existence of such a unit was proved in lemma 4.2; no unproved appeal to the classical Pell theorem remains.

6 Proof of the main theorem

Proof of theorem 1.1. Let (t, y) be any of the infinitely many integer pairs supplied by corollary 5.1. Define $P = P(t)$, $Q = Q(t)$, $R = R(t)$, and $S = S(t)$ by equations (8)–(11). By proposition 3.1,

$$2PQR + (P^2 - Q^2)S = 1 \quad \text{and} \quad P^2 + Q^2 - R^2 - S^2 = F(t) = y^2.$$

Therefore the hypotheses of lemma 2.1 hold. Define

$$x(t) = -(P(t)^2 + Q(t)^2), \tag{39}$$
$$z(t) = (P(t)^2 - Q(t)^2)R(t) - 2P(t)Q(t)S(t). \tag{40}$$

For reference, the latter polynomial expands to

$$z(t) = 50421655389668t^3 + 74872031013786t^2 + 37059612550518t + 6114499038718. \tag{41}$$

Then

$$x(t)^3 + x(t)^2y^2 + z(t)^2 + 1 = 0.$$

In fact, the same calculation before specializing $F(t)$ to the square y^2 gives the polynomial identity

$$x(t)^3 + x(t)^2F(t) + z(t)^2 + 1 = 0 \qquad \text{in } \mathbb{Z}[t]. \tag{42}$$

It remains only to verify that infinitely many distinct triples arise. From the coefficient calculation in proposition 3.1,

$$P(t)^2 + Q(t)^2 = 2357694538t^2 + 2333988934t + 577630421. \tag{43}$$

This quadratic has positive leading coefficient, so it tends to $+\infty$ as $|t| \to \infty$. By corollary 5.1, the admissible values of t are unbounded in absolute value. Consequently, (43) assumes infinitely many values on those admissible integers, and hence the integers $x(t)$ in (39) assume infinitely many distinct values. The resulting solution triples are therefore pairwise distinct along an infinite subsequence. $\square$

7 Concrete solutions and independent checks

Taking the initial Pell point $t = -659$ gives

$$\begin{aligned} P(-659) &= -20734347, & Q(-659) &= 24340322, \\ R(-659) &= 2238918, & S(-659) &= -13903675, \end{aligned}$$

and (37) gives $y = \pm 28706088$. The definitions (39) and (40) yield

$$\begin{aligned} x &= -1022364420580093, \\ z &= -1439774191877026309335 0. \end{aligned}$$

Thus, for example,

$$\boxed{(x, y, z) = (-1022364420580093,\ 28706088,\ -14397741918770263093350)} \tag{44}$$

is an integer solution. Its validity follows from the proved identities, but it may also be checked directly by exact integer arithmetic.

There is also a much smaller solution not needed for the infinite-family proof:

$$(x, y, z) = (-493, 12, 9210). \tag{45}$$

Indeed,

$$493 - 12^2 = 349 \quad \text{and} \quad 9210^2 + 1 = 84824101 = 493^2 \cdot 349.$$

Hence

$$(-493)^3 + (-493)^2 \cdot 12^2 + 9210^2 + 1 = 0.$$

By sign symmetry, both signs of 12 and both signs of 9210 give solutions.

8 Conclusion

The proof supplies an unconditional infinite construction. The essential points are:

1. the two-square identity in lemma 2.1, which converts the conditions (3) and (4) into a solution of the original equation;
2. the exact linear-polynomial identities in proposition 3.1; and
3. the self-contained Pell propagation theorem, which forces the quadratic $F(t)$ to be a square for infinitely many unbounded integer values of t.

Therefore (1) has infinitely many integer solutions, as claimed.